

Characteristics of Eye-tracking Metrics Obtained Using a Portable Device for Assessing Return to Duty Clearance After Concussion

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BACKGROUND

Assessing individuals for clearance for unrestricted return to activity following a concussion has been challenging due to a lack of objective tools. Standard practice relies on evaluating self-reported concussion symptom resolution in combination with balance and cognitive function. A pre-injury baseline test is often needed to help make return to duty assessments, but may not be available. A rapid, objective, baseline-free tool is needed. This research aimed to evaluate the sensitivity and specificity of a novel eye-tracking device within 72 hours of concussion and at the time of unrestricted return to duty clearance in military service academy (MSA) cadets.

METHODS

We recruited 84 concussed participants (33.3% Female) ages 17-23 years (mean 19.5, SD 1.1) from a MSA. Eye movements and symptom severity were collected on a handheld binocular-like device within 72 hours of concussion and at clinical clearance for unrestricted return to duty. The eye-tracker used a 167 Hz infrared camera during viewing of a 90-second video. An algorithm based on disrupted eye-tracking metrics and symptom severity after head injury was developed and validated. The performance of this algorithm was assessed for correlation with determination of clearance for return to activity.



RESULTS

Average eye movement score within 72 hours of injury was 5.9 (SD 3.1) vs. 5.2 (SD 3.0) for participants cleared for unrestricted return to activity. Average number of symptoms was 11.2 ± 5 within 72 hours of injury vs. 0.2 ± 1 for participants cleared for unrestricted return to activity. A model based on both eye movement metrics and symptom severity was 86.9% sensitive for concussion within 72 hours of injury (95% confidence interval (CI) 77.8 – 93.3%) and 97.6% specific (95% CI 91.7 – 99.7%) in correlating with patients recovered from concussion and cleared for unrestricted return to activity. The proportion classified as concussed by the model decreased over time from 86.9% at 72 hours post-injury to 2.4% at clearance for unrestricted return to activity (McNemar's $\chi^2(1) = 71.00$, $p < .001$).

DISCUSSION

A rapid, portable, baseline-free device demonstrated high sensitivity and specificity in assessing readiness for clearance for return to duty. The device has potential to support return to activity clearance in high-risk military populations and service academy cadets, using both subjective self-reported symptoms and objective eye-tracking metrics.

LEARNING OBJECTIVES

1. Describe how eye movements and symptoms change from acute injury to return to activity.
2. Assess the performance of an eye-tracking based model in predicting concussion classification after acute injury.
3. Assess the performance of an eye-tracking based model to support clearance for unrestricted return to duty after concussion in military service academy cadets.